

CE 613: Mechanics of Unsaturated Soils

Course Code	CE 613
Course Name	Mechanics of Unsaturated Soils
L-T-P-C	3-0-0-3
Prerequisites	Geotechnical Engineering I (CE302), Geotechnical Engineering II (CE 402) or equivalent course.
Intended for	UG and PG students
Mutual Exclusion	None
Approval	50th BoA (Jul 2023)

Course Contents

- **Introduction to Unsaturated Soil Mechanics:** Role of climatic conditions, Need and application areas of unsaturated soil mechanics, Typical profile of unsaturated soils (2 Hours)
- **Phase Properties and Stress state variables:** Properties of individual phases, Interaction of air and water, Volume-mass relations, Effective stress for unsaturated soils, Stress state variables, Limiting stress state conditions and experimental testing of stress state variables (8 Hours)
- **Measurement of soil suction:** Theory of soil suction, Capillarity, Measurement of Total suction, Matric suction, and Osmotic suction (8 Hours)
- **Flow behaviour in unsaturated soils:** Flow of water, Driving potential for water phase, Darcy's law, coefficient of permeability with respect to water phase, steady state flow (8 Hours)
- **Shear strength of unsaturated soils:** Failure envelope for unsaturated soils, Triaxial and direct shear tests on unsaturated soils (8 Hours)
- **Stress-deformation analysis for unsaturated soils:** Swelling pressure determination, 1-D Heave estimation, Foundation design in expansive soil (6 Hours)

Text Books

1. Fredlund, D.G., Rahardjo, H., and Fredlund, M.D., **Unsaturated Soil Mechanics in Engineering Practice**, 2nd Edition, John Wiley & Sons, Inc., 2012.
2. Lu, N., and Likos, W.J., **Unsaturated Soil Mechanics**, 1st Edition, John Wiley & Sons, Inc., 2004.

References

1. Ng, Charles, W.W., and Menzies, B., **Advanced Unsaturated Soil Mechanics and Engineering**, Taylor and Francis, 2007.
2. Blight, G.E., **Unsaturated Soil Mechanics in Geotechnical Practice**, Taylor and Francis, 2013
3. Refereed publications in the field of unsaturated soil mechanics